TRIAL TEST 1: ATOMIC STRUCTURE

Time allowed:	**70 minutes**	**Section 1** – Multiple Choice	20 marks
Total marks:	**80**	**Section 2** – Short & Extended Answer	60 marks

SECTION 1 – MULTIPLE CHOICE (20 MARKS)

1. The mass number of an atom is:
   (a) equal to the number of particles in the nucleus of that atom.
   (b) unique and identifies what element the atom is.
   (c) always greater than the mass number of the element that immediately precedes it in the periodic table.
   (d) the same as the mass number of all other isotopes of that atom.

2. A species has 4 neutrons, 2 electrons and 3 protons. This species could be:
   (a) An ion of an isotope of beryllium.
   (b) A neutral atom of an isotope of boron.
   (c) An ion of an isotope of lithium.
   (d) An ion of an isotope of carbon.

3. Which of the following is the correct electron configuration for the ion Al^{3+}?

   (a) 2,3.
   (b) 2,5.
   (c) 2,8.
   (d) 2,8,3.

4. Which of the following has the greatest number of electrons?

   (a) ${}^{14}_{7}N^{3-}$ (b) ${}^{19}_{9}F^{-}$ (c) ${}^{27}_{13}Al^{3+}$ (d) ${}^{23}_{11}Na$

5. Which of the following are isotopes of the same element?

Atom P:	36 protons	38 neutrons	37 electrons
Atom Q:	37 protons	38 neutrons	36 electrons
Atom R:	A = 76	Z = 38	
Atom S:	A = 80	its neutral atom contains 37 electrons	

   (a) Q and S
   (b) Q and R
   (c) P and S
   (d) R and S

6. Which of the following are listed in increasing order of ionisation energy?
   (a) Li, Na, K
   (b) F, O, N
   (c) Ne, Ar, Kr
   (d) B, C, N

7. An atoms electron is excited from its ground state (n=1) to the third energy level (n=3). As the electron returns to its ground state it gives up energy in the form of photons. How many different photon energies are possible in this case?
(a) 2
(b) 3
(c) 4
(d) 6

8. A mass spectrometer was used to determine the relative concentration of the three isotopes of an element X. The mass spectrum for the element contained three distinctive peaks with the following approximate heights; ^{24}X was 8.0 cm, ^{25}X was 1.0 cm and ^{26}X also 1.0 cm. From this information it can be concluded that the relative atomic mass (A_r) of element X is closest to:
(a) 24.0
(b) 24.5
(c) 25.0
(d) 26

9. Flame tests are carried out by placing small samples of metal salts in a Bunsen flame and observing the colours produced. Which of the following is correct regarding flame tests?
(a) The sample is placed in the flame using clean copper wire.
(b) Sodium salts burn with a bright green flame.
(c) The colours are due to electron transitions in the atomised sample.
(d) Flame tests can be used to identify the majority of metal ions.

10. Which of the following is not a feature of the atomic absorption spectrometer (AAS)?
(a) A bright Bunsen flame.
(b) A hollow cathode lamp.
(c) A source of pulsed light.
(d) A prism and monochromator.

SECTION 2 – SHORT AND EXTENDED ANSWER (60 MARKS)

Answer each question in the space provided.

11. (a) One of the isotopes of chlorine is symbolised as $^{37}_{17}Cl$. This indicates that for this isotope:

(i) the mass number is ________________

(ii) the atomic number is ________________

(iii) the number of neutrons in the nucleus is ________________

(iv) the number of electrons in a neutral atom would be ____________

(b) The electron configuration for sodium can be written as 2,8,1. Similarly, give the electron configuration for each of the following:

(i) a nitrogen atom ________________________________

(ii) a magnesium ion ________________________________

(iii) a sulfide ion ________________________________

(iv) a potassium atom ________________________________

[8 marks]

12. (a) What are valence electrons?

____

(b) State the number of valence electrons that each of the following atoms have.

(i) Be ______ (ii) Ar______ (iii) Na ______ (iv) Al ______

[6 marks]

13. The structure of the periodic table is closely linked to the electron configuration of the elements and is helpful in predicting their likely properties.

(a) What is common to the valence electrons of all elements in the third period of the periodic table?

____

(b) What is common about the number of valence electrons of all alkali elements?

____

(c) Why do elements belonging to the same group tend to have very similar chemical properties?

____

____

(d) Briefly explain why Group 18 elements, the noble or inert gases, are all unreactive.

____

____

(e) What link do chemists make between the electron structure of the inert gases and the electron interactions of all other elements?

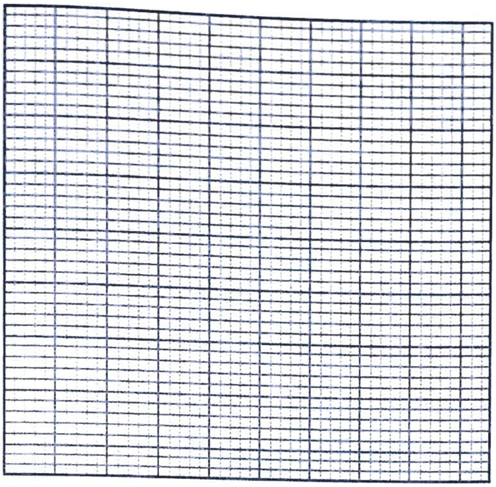
____

____

[10 marks]

14. Data determined from the mass spectrum for lead is shown below.

(a) Sketch the mass spectrum you would expect for these isotopes of lead on the blank graph grid below.



(b) Calculate the relative atomic mass (Ar) for the element lead.

[12 marks]

15. The electron of an atom has moved from its ground state (n = 1) to an excited state (n = 4) as shown in the diagram below.

(a) Has energy being absorbed or emitted by the atom?

(b) The excited electron can return to its ground state by one or several different steps. Indicate each of the possible transitions on the diagram above.

(c) Which transition would involve the largest wavelength photon?

(d) For this atom the transitions n = 4 to n = 2 and n = 3 to n = 2 both result in photons of visible light. If the two colours are red and green which transition would have resulted in the green?

[12 marks]

16. The atomic absorption spectrometer (AAS) developed in Australia in the 1950s, is a very useful and sensitive instrument for the analysis of small traces of metal contaminants. Briefly explain each of the following.

(a) Why the sample being analysed is placed in an intense flame.

(b) Why the light from the cathode lamp is passed through the flame.

(c) How the detector can distinguish light from the lamp from that emitted by the flame.

(d) What measurement indicates the presence, if any, and concentration of a target metal in the sample.

[12 marks]

END OF TEST (80 MARKS)

TRIAL TEST 2: MATERIALS AND BONDING

Time allowed: 60 minutes	**Section 1** – Multiple Choice 20 marks
Total marks: 80	**Section 2** – Short & Extended Answer 60 marks

SECTION 1 – MULTIPLE CHOICE (20 MARKS)

1. Which of the following contains only pure substances?
   (a) water, zinc, copper (II) sulfate, iron (II) oxide
   (b) neon, sodium chloride solution, sugar solution, argon
   (c) gold, magnesium oxide, petrol, chlorine
   (d) copper, aluminium oxide, oxygen, air

2. Which of the following is INCORRECT?
   (a) Elements and compounds are pure substances.
   (b) Mixtures can be separated by physical means.
   (c) Mixtures may be homogeneous or heterogeneous.
   (d) Sublimation refers to the phase change of a solid to a liquid.

3. In order to separate and recover sugar crystals from sand the order of the different steps is most likely to be:
   (a) dissolution, evaporation, filtration, crystallisation
   (b) dissolution, filtration, evaporation, crystallisation
   (c) dissolution, distillation, crystallisation, evaporation
   (d) filtration, evaporation, crystallisation.

4. Nanoparticles are defined as being those between:
   (a) 1.0 nm and 1.0 μm
   (b) 1.0×10^{-9} m and 1.0×10^{-10} m
   (c) 1.0×10^{-8} m and 1.0×10^{-9} m
   (d) 1.0×10^{-7} m and 1.0×10^{-9} m

5. Nanoparticles exhibit markedly different properties to those of their larger counterparts in bulk materials. The most likely reason for this is:
   (a) they are all made up of covalently bonded carbon atoms
   (b) they all have a large surface area to volume ratio
   (c) they all have a very low aspect ratio
   (d) they are all smaller than the wavelength of visible light.

6. Which of the following statements is INCORRECT for an ionic solid?
   (a) The substance will have a high melting point because of the strong electrostatic attraction between oppositely charged ions.
   (b) When heated sufficiently charged particles can move more freely and allow the passage of an electric current through the substance.
   (c) When the ions in the crystal lattice are forced to move, electrostatic repulsion tends to make the solid shatter.
   (d) When dissolved in water, the ionic lattice breaks up and makes electrons available to allow the passage of an electric current through the solution.

7. Which of the following statements is correct?
   (a) Atoms from group 17 on the periodic table share electrons with atoms from group 1 to form a covalent bond.
   (b) Atoms with nearly empty outer energy levels tend to share valence electrons so as to get a full outer energy level.
   (c) Substances with covalent bonds have more electrons and are harder to melt.
   (d) Atoms with nearly full outer energy levels can share valence electrons with other similar atoms to attain a full outer energy level.

8. Which of the following statements about metals is INCORRECT?
   (a) Metals are malleable.
   (b) With few exceptions, metals are silvery grey in colour.
   (c) Due to the strength of the metallic bond, ions in the metallic lattice are very close giving all metals high densities (when compared to water).
   (d) An ionic substance can be formed by reacting a metallic substance with a non-metallic substance.

9. Which of the following statements is true about the allotropes of carbon?
   (a) All are good conductors due to delocalised electrons.
   (b) All contain covalently bonded carbon atoms.
   (c) All their carbon atoms are bonded to three other carbon atoms.
   (d) All their structures are fairly similar.

10. Two elements X and Y have the following electron configurations:

    X: 2 , 8 , 2 Y: 2 , 7

    Which of the following is the most likely result if the two elements combine?

    (a) An ionic substance with the formula XY_2
    (b) An ionic substance with the formula X_2Y
    (c) A covalent substance with the formula XY_2
    (d) A covalent substance with the formula X_2Y_7

SECTION 2 – SHORT AND EXTENDED ANSWER (60 MARKS)

Answer each question in the space provided.

11. (a) **Classify each of the following as either element, compound or mixture:**
sugar , steam , air , oxygen , ethanol , vinegar , copper (II) oxide , potassium , limestone

element: ______________________________

compound: ______________________________

mixture: ______________________________

(b) **Which of the above are:**

(i) **pure substances?** ______________________________

(ii) **heterogeneous mixtures?** ______________________________

(iii) **cannot be separated into simpler substances?** ______________________________

[6 marks]

12. Pure substances can be separated from mixtures by using physical separation techniques. Separation is possible due to the substances having different properties. Complete the table by naming the technique for each example and the property/s involved.

SUBSTANCE TO BE RECOVERED	SEPARATION TECHNIQUE/S	PROPERTY/S MAKING SEPARATION POSSIBLE
salt from a salt/sand/water mixture		
water from a salt/water mixture		
$KCl(s)$ from a $KCl(aq)$ + $CuCl_2(aq)$ mixture		
alcohol from a alcohol/ water mixture		
plant pigment from plant colouring		

[10 marks]

13. For each of the following name the nanoparticle/s used and briefly explain their effect.

(a) Sunscreen lotion ______________________________

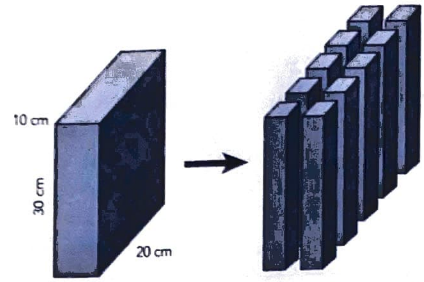
____

(b) Antibacterial bandages ______________________________

____

[6 marks]

14. A rectangular block of firewood is shown below. It measures approximately 30 cm high, 20 cm wide and 10cm thick. The block is split into ten equal pieces of 30 cm by 5.0 cm by 4.0 cm.



Wood block 30 cm x 20 cm x 10 cm   Ten pieces each 30 cm x 5 cm x 4 cm

(a) Assuming no loss of wood when it is split determine,

(i) The initial surface area of the block

___

___

(ii) The final total surface area of the ten pieces

___

___

(iii) The factor by which the surface area changed

___

___

(b) It was found that the small pieces of wood burn much more quickly than a large block. In terms of your results above explain clearly why this is the case.

___

___

[10 marks]

15. Covalent bonds form readily between atoms that have almost full valence shells. An example is the formation of a fluorine molecule from two fluorine atoms.

(a) Use electron dot diagrams to illustrate and explain how this formation occurs.

(b) Explain why a bond forms between the two atoms.

(c) Use the nature of this type of bond to explain why molecular solids are poor conductors of electricity.

(d) The type of bond that would form between fluorine and sodium atoms would be of quite a different nature. What type of bond would form and give a reason?

[12 marks]

16. Classify the following substances by placing them into their correct classification in the table below: *ammonia, sulfur dioxide, silicon dioxide, calcium oxide, potassium, lithium bromide, water, diamond.*

METALLIC	IONIC	COVALENT MOLECULAR	COVALENT NETWORK

[4 marks]

17. A solid ionic compound consists of an infinite array of oppositely charged ions. A typical crystal lattice, such as that of sodium chloride is shown. With the aid of diagrams and electron dot diagrams where necessary clearly explain:

(a) the formation of an ionic bond between sodium and chlorine atoms.

(b) why ionic solids are hard and brittle.

(c) why ionic solids do not conduct electricity.

[12 marks]

END OF TEST (80 MARKS)

SECTION 2 – SHORT AND EXTENDED ANSWER (60 MARKS)

Answer each question in the space provided.

11. (a) **The relative atomic mass (A_r) for helium is 4.00 and for oxygen is 16.0. What specifically does this information tell us about the atoms of these elements?**

(b) **Calculate the relative molecular mass for sulfuric acid and hydrochloric acid.**

(c) Which of the acids above has the greatest percentage by mass of hydrogen? What is this percentage?

[8 marks]

12. For each of the following determine the number of:

(a) moles of gold atoms in 6.022×10^{25} atoms of gold

(b) lead atoms in 0.25 mol of lead

(c) molecules of water in 2.5 mol of water

(d) molecules of sulfur dioxide in 64.0 g of this gas

TRIAL TEST 6: INTERMOLECULAR FORCES

Time allowed:	**70 minutes**	**Section 1** – Multiple Choice	20 marks
Total marks:	**80**	**Section 2** – Short & Extended Answer	60 marks

SECTION 1 – MULTIPLE CHOICE (20 MARKS)

1. Which of the following compounds would contain polar covalent bonds?
   (a) NaCl
   (b) S_8
   (c) Fe_2O_3
   (d) NH_3

2. Choose the list below which shows the bonds arranged in order of increasing polarity.
   (a) H-H, H-N, H-F
   (b) N-H, P-H, As-H
   (c) H-Cl, H-C, H-Cl
   (d) Cl-Cl, H-C, H-Cl

3. Which of the following lists contains molecules that are likely to be the most soluble in water?
   (a) HCl, C_3H_8, N_2
   (b) Cl_2, H_2, CCl_4
   (c) H_2S, NH_3, HF
   (d) HBr, C_6H_{14}, F_2

4. Which of the following statements about chromatography is incorrect?
   (a) Chromatography can be used to separate and identify very small quantities of pollutants in river water.
   (b) In gas chromatography the molecules in the mixture are separated according to the strength of intermolecular forces between and the different molecules in the stationary phase.
   (c) In high performance liquid chromatography the molecules in the mobile phase must be strongly attracted to molecules in the stationary phase to ensure complete separation and identification of the molecules in the mobile phase.
   (d) Two reasons why thin layer chromatography is preferred over paper chromatography are sensitivity and destruction of cellulose by corrosive agents.

5. Which of the following statements is most correct?
   (a) $BeCl_2$ cannot exist as a molecule because Be does not have 8 electrons in its outermost shell after it has bonded with two chlorine atoms.
   (b) NH_3 is a polar molecule that is pyramidal in shape as it has one lone pair of electrons.
   (c) CH_4 is a polar molecule because carbon and hydrogen have different electronegativities.
   (d) NH_3 is likely to be a less soluble in hexane than it is in water because hexane is a non-polar molecule.

6. Choose the row in the table below that correctly lists examples of the main intermolecular force exhibited by the covalent molecular substances listed.

	Hydrogen Bonding	Dipole-Dipole	Dispersion
(a)	H_2Te	NH_3	F_2
(b)	H_2S	HF	HCl
(c)	H_2O	HCl	C_2H_6
(d)	H_2O	CCl_4	CH_3CH_3

7. Which of the following covalent molecular compounds is both a polar molecule and tetrahedral in shape?

(a) CH_3Cl
(b) CCl_4
(c) H_2O
(d) CH_2CH_2

8. Ammonia has a boiling point of – 33 °C while phosphine (PH_3) has a boiling point of -88 °C. The difference between these boiling points is most likely due to the fact that:

(a) ammonia molecules form hydrogen bonds whereas phosphine molecules do not.
(b) the dispersion forces between phosphine molecules are stronger than those between ammonia molecules.
(c) the dispersion forces between ammonia molecules are stronger than those between phosphine molecules.
(d) phosphine molecules form hydrogen bonds whereas ammonia molecules do not.

9. Choose the group that is different to the other three.

(a)	CCl_4	H_3O^+	H_2
(b)	CH_4	NH_4^+	H_2S
(c)	NH_4^+	NH_3	HCl
(d)	CF_4	PCl_3	BeF_2

10. Choose the group that has the pure substances correctly listed in order of increasing melting point.

(a)	NH_3	CH_4	SiO_2	W
(b)	Cl_2	CO_2	Na	C_7H_{16}
(c)	CCl_4	$CaCO_3$	$Al_2(CO_3)_3$	H_2CO_3
(d)	NH_3	C_7H_{16}	Na	SiO_2

SECTION 2 – SHORT AND EXTENDED ANSWER (60 MARKS)

Answer each question in the space provided.

11. **Name a substance whose properties match those described.**

(a) **A highly polar molecular substance that is very soluble in water. Individual molecules are described as having a pyramidal shape.**

____

(b) **Combines with another element sharing one pair of valence electrons. When combined with hydrogen it forms molecules that exhibit hydrogen bonding and are acidic.**

____

(c) **Has a very high first ionisation energy, a very low boiling point and does not form bonds with the two elements that have atomic numbers either one smaller or one greater than its own.**

____

[6 marks]

12. Draw the electron dot diagrams (Lewis structures) for the following:

(a) Cl_2

(b) NH_3

(c) CO_2

(d) NO_3^-

(e) SO_4^{2-}

(f) C_2H_4

[12 marks]

13. Complete the following table by stating the shape and polarity of each molecule

NAME	SHAPE	POLAR OR NON-POLAR?
chloroform ($CHCl_3$)		
hydrogen sulfide		
phosphorus trichloride		
dichlorine monoxide		

[8 marks]

14. Explain why H_2O has a higher boiling point than H_2S but H_2S has a lower boiling point than H_2Se.

[6 marks]

15. Explain the following trend in boiling points: $CH_4 < C_2H_6 < C_3H_8$

[6 marks]

16. Explain why polar molecular compounds are often soluble in water but non-polar molecular compounds are rarely soluble in water. Include a discussion on the intermolecular processes involved in dissolving.

[6 marks]

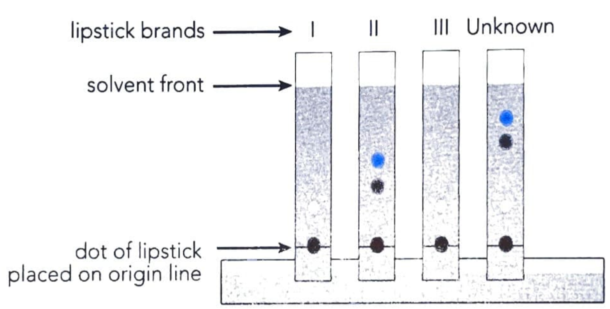
17. (a) **Explain why CO is a polar molecule while CO_2 is a non-polar molecule.**

[4 marks]

(b) **At a temperature of -100° C and a pressure of 100 kPa, CO is a gas and CO_2 is a solid. In terms of intermolecular forces, suggest a reason why the non-polar CO_2 is a solid while the polar CO is a gas.**

[2 marks]

18. A student was given the task to analyse three brands of lipstick to see which one matched an unknown brand. The student conducted a paper chromatography analysis as summarised below.



The mobile phase used was a 50% ethanol/50% acetone mixture.

(a) Explain why the origin line for placing the dot of lipstick was drawn in pencil and not pen.

[2 marks]

(b) Suggest why the mobile phase was an ethanol/acetone mixture rather that water.

[2 marks]

(c) Explain why the different colours (compounds) travel to different heights up the chromatography paper.

[2 marks]

(d) Which of the lipstick brands is the unknown brand likely to be? Give a brief justification for your answer.

[2 marks]

(e) In the forensic analysis of these lipsticks, HPLC is preferred over gas chromatography. What does this suggest about the boiling points of the coloured compounds being tested.

[2 marks]

END OF TEST (80 MARKS)

TRIAL TEST 8: SOLUTIONS AND SOLUBILITY

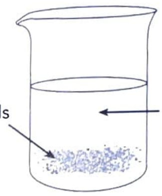
Time allowed:	**70 minutes**	**Section 1** – Multiple Choice	20 marks
Total marks:	**80**	**Section 2** – Short & Extended Answer	60 marks

SECTION 1 – MULTIPLE CHOICE (20 MARKS)

1. A solution may best be described as:
   (a) a pure substance of constant composition.
   (b) a homogeneous mixture of uniform composition.
   (c) a substance which can be purified by filtration.
   (d) a heterogeneous mixture of variable composition.

2. Gases most easily dissolve in water at:
   (a) low temperature and low pressure.
   (b) low temperature and high pressure.
   (c) high temperature and low pressure.
   (d) high temperature and high pressure.

3. A quantity of sodium chloride is added to water in a beaker and stirred briskly for several minutes. The mixture is allowed to settle and appears as a clear colourless liquid with some white crystals at the bottom as illustrated. It would be true to say that:



   (a) the clear liquid is a saturated solution.
   (b) a little more water is needed in order to produce a saturated solution.
   (c) the clear liquid is a supersaturated solution.
   (d) cooling the solution would make it more concentrated.

4. Adding some salt to water will affect its properties. It would be true to say that the salt would:
   (a) raise the water's vapour pressure.
   (b) raise the water's boiling point.
   (c) raise the water's freezing point.
   (d) lower the water's boiling point.

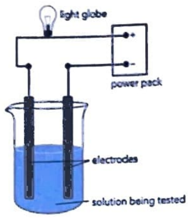
5. Approximately equal volumes of the two solutions in the following pairs of solutions listed below are mixed together. All solutions are 0.1 mol L^{-1}. A precipitate will form in:
   (a) (i) and (ii) only.
   (b) (ii) and (iii) only.
   (c) (i), (ii) and (iii) only.
   (d) (i), (ii) and (iv) only.

(i)	$Pb(NO_3)_{2(aq)}$	and	$Na_2CO_{3(aq)}$
(ii)	$NaCl_{(aq)}$	and	$AgNO_{3(aq)}$
(iii)	$BaCl_{2(aq)}$	and	$Na_2SO_{4(aq)}$
(iv)	$NH_4NO_{3(aq)}$	and	$KCl_{(aq)}$
(v)	$MgCl_{2(aq)}$	and	$(NH_4)_2SO_{4(aq)}$

6. Which of the following equations correctly expresses the dissolving of barium chloride in water?

(a) $BaCl_{2(s)} + H_2O \rightarrow BaCl_{2(aq)}$
(b) $BaCl_{2(s)} \rightarrow Ba^{2+}_{(aq)} + 2Cl^-_{(aq)}$
(c) $BaCl_{2(s)} + H_2O \rightarrow BaCl_{(aq)} + Cl^-_{(aq)}$
(d) $BaCl_{2(s)} \rightarrow BaCl_{2(aq)}$

7. Conductivity tests were carried out using the apparatus shown at right.



Test A - beaker contains distilled water. When hydrogen chloride gas is bubbled through it the globe begins to glow brightly.

Test B - beaker contains ethanol. When hydrogen chloride gas is bubbled through it the globe does not glow.

Which of the following best explains these observations?

(a) Electrons are able to flow through an aqueous solution of $HCl_{(g)}$ but not through an ethanol solution of $HCl_{(g)}$.
(b) Distilled water is a good conductor of electricity whereas ethanol is not.
(c) Water molecules are more easily ionised than ethanol molecules.
(d) $HCl_{(g)}$ dissolves in water to form charged particles whereas in ethanol it remains as neutral molecules.

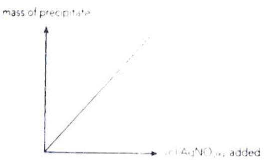
8. Conductivity apparatus similar to that in question 7 is used to test the conductivity of the following solutions. In which case will the globe glow brightest?

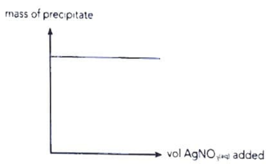
(a) 300 mL of 1.0 mol L^{-1} $HCl_{(aq)}$
(b) 100 mL of 2.0 mol L^{-1} $H_2CO_{3(aq)}$
(c) 300 mL of 1.0 mol L^{-1} $H_2CO_{3(aq)}$
(d) 200 mL of 2.0 mol L^{-1} $CH_3COOH_{(aq)}$

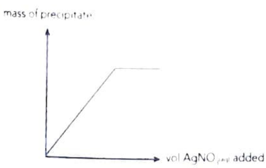
9. If 2.0 mg of sodium chloride crystals were dissolved in 100 mL of distilled water which of the following would most correctly describe the solution? Assume density of the resulting solution to be 1.0 g mL^{-1}.

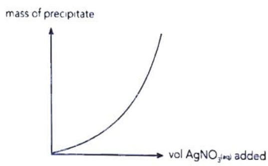
(a) A 2.0% salt solution.
(b) A 20 g L^{-1} salt solution.
(c) A 0.348 mol L^{-1} salt solution.
(d) A 20 ppm salt solution.

10. Silver nitrate solution was added in excess to a sodium chloride solution. The mixture was constantly stirred as the silver nitrate solution was added. Select which of the following graphs most likely represents the mass of precipitate formed as the mixing occurred.

(a) 

(b) 

(c) 

(d) 

SECTION 2 – SHORT AND EXTENDED ANSWER (60 MARKS)

Answer each question in the space provided beneath the question. Show all working where calculations are involved.

11. **(a)** **Dissociation and ionisation are different processes. Explain what occurs in each case.**

(i) **Dissociation**

(ii) **Ionisation**

[4 marks]

(b) Give a balanced equation to illustrate each of the above processes.

(i) Dissociation

(ii) Ionisation

[4 marks]

12. Write balanced **ionic** equations and describe what would be observed when:

(a) sodium sulfate solution is mixed with barium hydroxide solution.

equation:

observation:

(b) some calcium hydroxide powder is added to water.

equation:

observation:

(c) a few drops of silver nitrate are added to household tap water.

equation:

observation:

[9 marks]

13. (a) **Briefly describe what is meant by an electrolyte.**

(b) **List each of the following as either strong, weak or non-electrolytes.**

$C_{12}H_{22}O_{11}$ (sugar), K_2CO_3, $NaCH_3COO$, H_3PO_4, $MgSO_4$,
H_2O, NH_3, H_2CO_3, CH_3CH_2OH (ethanol)

(i) strong electrolytes: ______

(ii) weak electrolytes: ______

(iii) non-electrolytes: ______

[5 marks]

14. Clearly explain each of the following:

(a) What is meant by a saturated solution.

(b) Why salty water is more difficult to freeze than pure water.

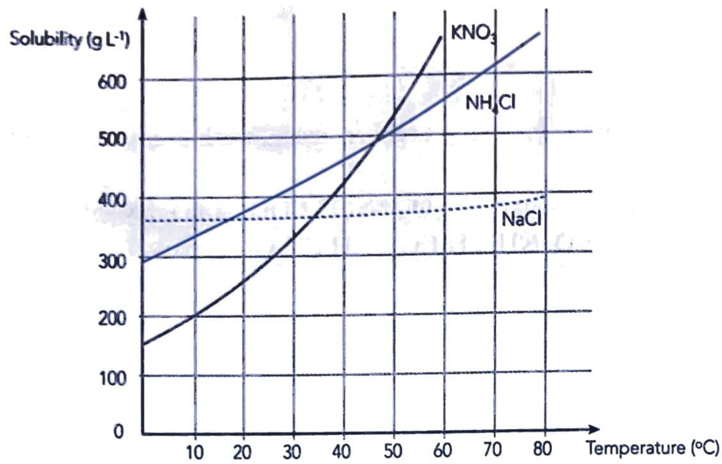
(c) Why a salt solution can conduct an electric current but a sugar solution cannot.

(d) Why a 1.0 mol L^{-1} solution of $MgCl_2$ is a better conductor than a 1.0 mol L^{-1} solution of NH_4Cl.

(e) Why the addition of salt to water reduces its vapour pressure and increases its boiling point.

[10 marks]

15. Use the graph of solubility shown to answer the following. Show all working.



(a) What is the maximum amount of NaCl that can be dissolved in 5.0 L of NaCl solution at 20°C ?

___

___

___

(b) 100 g of NH_4Cl were dissolved in warm water to make up 250 mL of solution. To what temperature must this solution be cooled for crystals to form?

___

___

___

(c) 100 g of KNO_3 crystals are dissolved in distilled water at 50°C to make up 200 mL of solution. The solution is then allowed to cool to 10°C. What mass of KNO_3 crystals will form?

___

___

___

[12 marks]

16. (a) A solution of barium chloride is made up by dissolving 2.55 g of this substance in distilled water and making up to 250.0 mL. Calculate the concentration of this solution in:

(i) grams per litre

___

___

(ii) moles per litre

[6 marks]

(b) A particular brand of beer contains 5.0% (w/w) alcohol (C_2H_5OH). Calculate the mass of alcohol that would be contained in a 330 mL bottle of this beer.

[4 marks]

(c) A 5.00 kg sample of sea water was evaporated to dryness and 155.5 g of solids remained.

Calculate the concentration of the solids in:

(i) % by mass

(ii) ppm

[6 marks]

END OF TEST (80 MARKS)